

Yanjin Yang

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About

Yanjin Yang is currently a fourth-year direct-admission Ph.D. student. His research interests lie in source code security and API security. In software code security, he studies LLM agent-based approaches for vulnerability detection and evaluates the generalization and robustness of the underlying LLM models in the vulnerability detection task. Besides, he also analyzes how external vulnerability reports drive code changes across the software supply chain. In API security, he focuses on leveraging agents to examine API entry points and calling processes for security vulnerabilities.

Education

- **Nanjing University** Nanjing, China
direct-admission Ph.D. student in Software Engineering (Headmaster Special Scholarship) Sep. 2022 - Present
- **Chongqing University** Chongqing, China
B.Sc. in Software Engineering (Outstanding Undergraduate: GPA 3.88/4) Sep. 2018 - July 2022

Research

- [1] **Yang Y**, Zhou X, Mao R, Xu J, Yang L, Zhang Y, Shen H, Zhang H. Dlap: A deep learning augmented large language model prompting framework for software vulnerability detection. *Journal of Systems and Software*, 2025.
- [2] **Yang Y**, Liu B, Zhang H, Xu J, Zhou X, Lyu J, Mao Z, Pan D. Towards Robust Detection for Malicious Injection Variants. *IEEE Transection on Software Engineering*, (Under Revision 2).
- [3] **Yang Y**, Mao R, Tan R, Shen H, Rong G. Robustness Verification Method for Artificial Intelligence Systems Based on Source Code Processing. *Journal of Software (Chinese)*, 2023
- [4] **Yang Y**, Gao M, Li Y, Wu F, Wang J, Zhao Q. MSPLD: Shilling Attack Detection Model Based on Meta Self-Paced Learning. *International Joint Conference on Neural Networks (Oral)*, 2021
- [5] Xu J, Zhang H, **Yang Y**, Yang L, Cheng Z, Lyu J, Liu B, Zhou X, Alberto B, Yin KC, Thiam KC. One Size Does Not Fit All: Investigating Efficacy of Perplexity in Detecting LLM-Generated Code. *ACM Transactions on Software Engineering and Methodology*, 2025
- [6] Xu J, Zhang H, Zhou X, **Yang Y**, Mao R, Li X, Yang L, Shen H. Automated detection of affected libraries from vulnerability reports. *Automated software engineering*, 2025
- [7] Cheng Z*, **Yang Y*(Equal Contribution)**, Zhang H, Yang L, Hu J, Xu J, Liu B, Shen H. Leveraging Self-Paced Learning for Software Vulnerability Detection. *ACM Transactions on Software Engineering and Methodology*, (Under Review)

Projects

Proj1. CCF Populus Grove Project: Multi-Agent Collaboration Task Error Detection (09.2025 - present)

We will conduct anomalies detection in multi-agent collaboration system during the multi-agent cooperation process in code generation. Correcting these anomalies and providing feedback enhances the ability of multi-agent to complete code generation tasks.

Proj2. CCF Populus Grove Project: Vulnerability Detection Based on LLM in OpenHarmony (09.2024 - 08.2025)

We fine-tuned LLM using open-source HarmonyOS vulnerability information to improve its capability in vulnerability detection tasks. I was responsible for the design and optimization of the detection model. Building on the existing research foundation of LLM-based vulnerability detection “DLAP”, the technical solution was further refined and adapted for enterprise-level requirements within the HarmonyOS ecosystem.

Proj3. CCF Populus Grove Project: PaaS API Security Guardrail (03.2024 - 08.2024)

We use Agentic AI to enhance API security by detecting malicious payloads and abnormal API call patterns. I designed all core algorithms, optimized key performance metrics, and served as the first author for all results. This work produced a black-box API discovery algorithm and effective methods for addressing payloads injection and broken access control risks from the OWASP API Security Top 10. The solutions were deployed in the Cloud PaaS intelligent gateway in practice, leading to two enterprise patents.

Patents

- [1] **Yang Y**, Liu B, Wang Z, Zhang H, Mao Z, Pan D. “A Detection Model Training Method, Apparatus, Device, and Storage Medium.” *Chinese Patent Application No. 20241072521-9.1*, (application accepted), 2024.
- [2] Mao Z, Pan D, Xia D, **Yang Y**, Liu B, Zhang H. “An Anomaly Detection Method, Training Method, and Related Apparatus.” *Chinese Patent Application No. 20251011227-7.1*, (application accepted), 2025.
- [3] Zhou X, **Yang Y**, Mao R, Zhang H, Xu J, Zhang Y. “A Deep-Learning-Enhanced Prompting Method for Large Language Models in Software Vulnerability Detection.” *Chinese Patent Application No. 20241060518-2.9*, (application accepted), 2024.
- [4] Zhou X, Tan R, Mao R, **Yang Y**, Zhang H, Zhang Y. “A Fine-Grained Vulnerability Localization Method Based on a Pretrained Model.” *Chinese Patent Application No. 202410608234-3.1*, (application accepted), 2024.
- [5] Zhou X, Zhang Y, **Yang Y**, Zhang H, Mao R, Xu J, Zhang Y. “Deep-Learning-Enhanced Large Language Model for Vulnerability Detection in Operating System Kernel Scenarios.” *Chinese Patent Application No. 20231172193-0.1*, (application accepted), 2023.

Selected Honors

- **Nanjing University Elite Scholarship** (2024, 2025)
- **Outstanding Graduate Student Representative** (2024)
- **China Optics Valley Scholarship** (2023)
- **Outstanding Graduate Student Leader** (2023)
- **Second Prize in the National Software Innovation Competition** (2022)

Work In Progress

- **Towards Robust Vulnerability Detection: Agentic-AI-Constructed Meta Data for Meta Self-Paced Noise-Robust Learning**
This research explores how to teach AI to detect software vulnerabilities in a way that mirrors human learning: starting from simpler, more reliable examples and gradually moving to harder ones. We use an Agentic AI to build a clean and trustworthy meta dataset, which serves as a guide for a Meta Self-Paced Learning process. With this guidance, the model learns to prioritize easy and correct samples first, then progressively handle more complex cases, improving its robustness to noisy real-world labels.